Indian Institute of Technology Gandhinagar

ES116 Principles and Applications of Electrical Engineering

Semester II, 2025-2026

Tutorial-8 (Op-AMP based functionalities)

Q1. You have a carbon-fibre strain gauge on a bridge that outputs a 1 V signal and has an internal resistance of 10 kΩ. You need to feed this signal into a digital controller that also has an input resistance of 10 kΩ.

A. If you connect the sensor directly to the controller, calculate the actual voltage the controller will read. What is the percentage of data lost?

B. You insert an ideal non-inverting Op-Amp buffer (Voltage Follower) between the sensor and the controller. Calculate the new voltage read by the controller, and prove mathematically how the buffer fixes the problem.

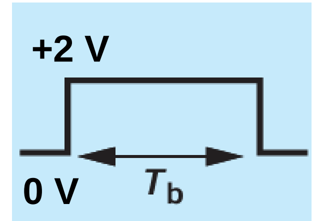
Q2. A drone needs to combine data from three sensors: a gyroscope (v_1), an optical flow sensor (v_2), and an accelerometer (v_3). The drone's microcontroller has a limited number of physical input pins. To save connection pins and make the computer's job simpler, you decide to mix these three analog signals into one single signal *before* plugging it into the computer. Design a circuit to physically calculate this exact equation: $\mathbf{V_{control} = -(2v_1 - 4v_2 + 0.5v_3)}$.

Q3. You are designing the closed-loop temperature control system for a plasma etching chamber in the TATA's semiconductor foundry. The silicon wafer's temperature is monitored by a precision differential sensor that outputs **a tiny 10 mV signal (V_d)**. However, the high-power RF generator used to strike the plasma is located right next to the sensor array. This generator induces a massive **5 V common-mode noise signal (V_{cm})** equally onto both of your delicate sensor wires. Your receiver circuit uses a difference amplifier with a built-in differential gain of $A_d = 100$.

A. If you use a basic Op-Amp with a Common-Mode Rejection Ratio (CMRR) of **40 dB**, calculate the exact common-mode gain (A_{cm}) and the total voltage seen at the output (Signal + Noise). Explain why this will ruin the wafer batch.

B. You upgrade the board to a precision instrumentation amplifier with a CMRR of **100 dB**. Calculate the new total voltage at the output and prove mathematically why the controller can now do its job.

Q4. You are designing a budget robot vacuum. To save money, the front bumper switches were removed. The only sensor left is a tachometer (speed monitor) on the wheels that outputs **+2V** when driving forward, and **-2V** when driving in reverse. You split this speed signal and feed it into two different Op-Amp circuits to help the robot navigate.



A. You feed the speed signal into **an Op-Amp Integrator**. What physical property of the robot (position, speed, or acceleration) does this output represent? Plot the output for the given input. Assume zero initial conditions across the capacitor.

B. You feed the exact same speed signal into **an Op-Amp Differentiator**. What physical property does this output represent? Plot the output for the given input. Assume zero initial conditions across the capacitor.

C. To design the resistor and capacitor values for these circuits, you are given following scenarios:

1. It takes the robot exactly 10 seconds to drive across the room. You want the **Integrator** to output exactly **-10V after 10 seconds** of driving forward at **+2V.** If you choose a standard capacitor of **C=10 μF**, calculate the required value of the resistor (R). Assume OpAMP DC Supplies of $\pm$ 12 V.

2. When the robot smashes into the wall, it violently switches from driving forward **(+2V**) to driving backward (**-2V**). This sudden change takes just **0.1 seconds**. You want the **Differentiator** to output a sharp **+5V spike** when this happens to act as a crash alarm for the computer. If you choose a capacitor of **C=1 μF**, calculate the required value of the resistor (R).

Q5. You are building an EEG monitor to read a patient's brainwaves. The target brainwave is a tiny, alternating signal at 10 Hz. However, you face the following two major signal problems:

- The metal sensors on the patient's skin produce a constant, **steady +0.1 V** (i.e. 0 Hz DC voltage).
- The long sensor cables act like antennas and pick up **high-frequency 60 Hz electrical noise** from the room's lighting.

A. You first build a basic Integrator using an input **resistor R_{in} =10 kΩ and a feedback capacitor C=100 nF**. What is the theoretical gain of this circuit at 0 Hz? What will eventually happen to the Op-Amp's output if it tries to integrate that constant +0.1 V skin voltage? Assume OpAMP open loop gain to **be 100 dB**.

B. To save the circuit, you add a feedback resistor **R_f =100 kΩ in parallel with the capacitor**. Calculate the newly stabilized output voltage caused by the **+0.1 V skin sensor.**

C. Plot the **Gain with frequency** for the modified circuit. Calculate **the cutoff frequency (f_c).** Does this cutoff frequency successfully allow the 10 Hz brainwave to pass through while blocking the 60 Hz high-frequency room noise?

(Optional, for brave hearts) Q6. Draw a block diagram (using Op-Amp symbols for Integrators and Inverting Amplifiers) to solve the following first-order differential equation for y(t): $dy/dt = -3y(t) + x(t)$

(Hint: Start by assuming you already have a wire carrying the signal dy/dt . What must you pass it through to get y(t)? How do you loop it back?)

<u>More Practice Problems</u>

- Behzad Razavi, Fundamentals of Microelectronics.
- Irwin J.D. and Nelms R.M., Basic Engineering Circuit Analysis.